

SAHO – Product Requirements Document

Phase 8 – Testing, Deployment & Competition Readiness

8.1 Objective

Define the quality assurance strategy, deployment process, repository standards, and competition readiness checklist for SAHO. This phase ensures the solution is reliable, maintainable, secure, and presentation-ready.

8.2 Testing Strategy

Testing Level	Scope
Unit Tests	Utilities, hooks, services, AI response parsers
Integration Tests	Authentication, AI workflows, Firestore interactions
End-to-End Tests	Complete recovery, caregiver, and vision journeys
Accessibility Tests	Keyboard navigation, ARIA, screen readers, WCAG 2.2 AA
Performance Tests	Lighthouse, Core Web Vitals, offline support

8.3 Quality Gates

Every pull request should pass: • ESLint • TypeScript compilation • Unit tests • Integration tests • Build verification • No critical security issues

8.4 CI/CD Pipeline

Suggested pipeline: 1. Install dependencies 2. Lint 3. Run tests 4. Build application 5. Deploy preview 6. Deploy production after approval

8.5 Deployment

Frontend: Vercel Backend: Firebase Authentication: Firebase Auth Database: Firestore AI: Gemini API Environment variables managed securely with platform secrets.

8.6 Repository Standards

Repository should include: • README.md • LICENSE • .env.example • docs/ • src/ • tests/ • CHANGELOG.md • CONTRIBUTING.md

8.7 README Requirements

The README should describe: • Problem statement • Product overview • Features • Technology stack • Setup instructions • Architecture • AI workflow • Screenshots • Future roadmap • License

8.8 Demo Flow

Recommended judge demonstration: 1. Introduce the substance-use challenge. 2. Launch SAHO NOW. 3. Demonstrate voice-first crisis support. 4. Show guided breathing. 5. Demonstrate caregiver teleprompter. 6. Show trigger vision. 7. Show substance identification. 8. Display recovery timeline and educational hub. 9. Explain security, accessibility, and offline support.

8.9 Competition Checklist

✓ Public GitHub repository ✓ Single branch ✓ Repository size below competition limit ✓ README completed ✓ Environment example included ✓ Documentation committed ✓ Tested build ✓ Working deployment

8.10 Future Roadmap

Future enhancements: • Wearable integration • Therapist portal • Community support • Predictive relapse insights • Multilingual voice support • Regional emergency services integration

8.11 Definition of Done

SAHO is considered complete when: • Core recovery and caregiver journeys function correctly. • AI interactions are safe and validated. • Accessibility goals are met. • Performance targets exceed Lighthouse 95. • Security controls are implemented. • Documentation is complete. • The application is deployable and demo-ready.